

Suwignyo Prasetyo

Data Engineer · Backend Developer · Metocean Practitioner

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SUMMARY

Data Engineer and Backend Developer building reliable production data pipelines, workflow orchestration, and operational systems. Experienced in Python, Prefect, Linux, Docker, object storage, and HPC, with geospatial/metocean domain expertise — currently operating BMKG Ocean Forecasting System (BMKG-OFS) production workflows at the Centre for Marine Meteorology.

EXPERIENCE

Indonesian Agency for Meteorology, Climatology, and Geophysics (BMKG) Jan 2023 – Present

Centre for Marine Meteorology · Jakarta, Indonesia · Hybrid

Metocean Data Engineer Sep 2023 – Present

- Operate production data pipelines for BMKG-OFS (INACAWO, INAWAVES, INAFLOWS) supporting real-time marine forecasting.
- Build ingestion, validation, transformation, post-processing, and dissemination workflows with Python, Prefect, Linux, Docker, and SLURM.
- Develop cloud-optimized climatological/hindcast archives with Zarr and S3-compatible storage (25+ years, 3-hourly, 388k grids).
- Design and benchmark WAVEWATCH III / SWAN backup forecasting on the HPC DRC, including SAT and MPI/OpenMP studies.
- Build Streamlit scientific apps for analysis/monitoring; contribute as Big Data & AI engineer on MMS modernization.
- Provide L2/L3 troubleshooting for forecasting systems, pipelines, and production services.

Marine Meteorologist Feb 2023 – Sep 2023

- Produced operational marine forecasts and verified observational/satellite/NWP data for maritime safety, fisheries, and offshore ops.
- Supported field campaigns (ADCP surveys, Jalacitra Expedition) and Fishermen Weather Field School capacity-building.

SELECTED PROJECTS

- **BMKG-OFS Analysis Platform** (2025) — Streamlit analysis/monitoring (klimatologi.pusmar.id).
- **BMKG-OFS DRC Backup** (2024) — Standalone wave-forecast backup with HPC performance benchmarking.
- **Zarr archive for BMKG-OFS** (2024) — Large-scale multidimensional model archive for scalable geospatial access.
- **INACAWO post-processing** (2024) — Parallel post-processing for the 3-way coupled model (Baron Weather).
- **BMKG-OFS bias correction** (2024) — MOS correction of significant wave height vs satellite altimetry.
- **Coastal flood ConvLSTM** (2024) — Semarang coastal-flood model; AI Datathon finalist (BMKG × KORIKA).

SKILLS

Data engineering: Python, ETL/ELT, Prefect, Xarray, Zarr, NetCDF, data QC, time-series
Platform & backend: Linux, Bash, Docker, AWS / object storage, SLURM, APIs, Grafana
Geospatial & metocean: WAVEWATCH III, SWAN, NWP, hindcast, model verification, marine forecasting
Languages: Indonesian & Javanese (native); English (IELTS 6.5, CEFR B2)

EDUCATION

State College of Meteorology Climatology and Geophysics (STMKG) Sep 2018 – Sep 2022

BASc, Meteorology · GPA 3.70 / 4.00 · honours (3rd in Department) · fully funded BMKG scholarship
Thesis: moistening of cold-surge air masses over the western Maritime Continent using ERA5 (2010–2019).

HONORS, RESEARCH & TRAINING

- **Trainer** — Satellite Altimetry for Operational Metocean Services (BMKG × OTGA), 2024 · **AI Datathon Finalist** (BMKG × KORIKA), 2024
- **Karya Cipta Madya** (STMKG, 2022) · **BMKG Scholarship** (2018–2022) · WW3 optimization with Eviden (2024) · Ship Safety Score Model (peer-reviewed, 2024)
- Haryanto et al. (2023), *JPPA* 13(1) · Prasetyo et al. (2022), *Jurnal Aplikasi Meteorologi* 1(1)
- HPC Scientific Computing — NPTEL (2025) · Model Validation & HPC — CLS France (2024) · Intro to HPC — BMKG · AWS Cloud & Gen AI · INACAWO Hindcast